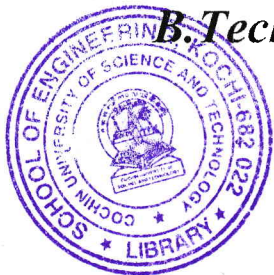


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B.Tech. Degree VIII Semester Supplementary Examination
August 2018

CS 1801 ADVANCED ARCHITECTURE AND PARALLEL PROCESSING
(2012 Scheme)

Time : 3 Hours

Maximum Marks : 100

PART A
 (Answer *ALL* questions)

(8 × 5 = 40)

- I. (a) Compare dataflow, program flow and reduction computers.
- (b) Distinguish between UMA and NUMA architecture models.
- (c) Illustrate cache memory addressing schemes.
- (d) Explain temporal locality, spatial locality and sequential locality associated with a memory hierarchy?
- (e) Discuss load-store forwarding and store-store forwarding.
- (f) Write short notes on linear and non-linear pipeline processors.
- (g) Differentiate between synchronous and asynchronous message passing?
- (h) Discuss different methods for vectorization.

PART B

(4 × 15 = 60)

- II. (a) Discuss Fynn's classification of parallel computer with suitable diagrams. (8)
- (b) Explain abstract model of parallel computer. (7)
- OR**
- III. (a) Describe in detail different types of dependencies in a program. (8)
- (b) Consider the following fragment of any program. (7)

S1 : Load R1, A
 S2 : Add R2, R1
 S3 : Move R1, R3
 S4 : Store B, R1

 Identify the dependencies and draw the dependency graph.
- IV. (a) Explain the shared memory organization. (8)
- (b) Discuss different bus arbitration schemes. (7)
- OR**
- V. Write notes on Superscalar, VLIW and scalar processors. (15)

(P.T.O.)

- VI. (a) Explain synchronous and asynchronous pipeline processors. (5)
(b) Discuss in detail cache coherence problem in data sharing (5)
(c) What is the principle of multi-threading? (5)
- OR**
- VII. (a) Explain the architecture of the following : (10)
(i) Crossbar switch (ii) multiport memory.
(b) Discuss about any one of SIMD computer model. (5)
- VIII. Explain in detail any three parallel programming models. (15)
- OR**
- IX. (a) Explain any two types of code optimization. (10)
(b) What is meant by instruction level parallelism? (5)
